

Multi-Server Java Application Build, Store, and Deployment System

Objective

The objective of this task is to set up a multi-server environment for building, storing, and deploying a Java web application using a Build Server, Nexus Repository Server, and Deploy Server.

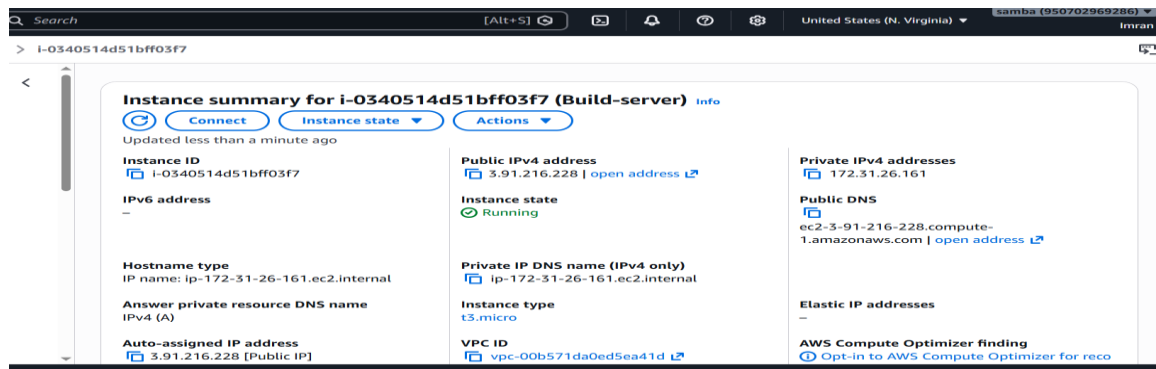
Servers Used

1. Build Server – Build Java application using Maven
2. Nexus Repository Server – Store WAR artifacts
3. Deploy Server – Deploy application on Apache Tomcat

1. Java Build Server Setup

Step 1: Connect to Build Server

```
ssh -i "devops.pem" ubuntu@ec2-3-91-216-228.compute-1.amazonaws.com
```



Step 2: Install Java and Maven

```
sudo apt update
```

```
sudo apt install openjdk-17-jdk -y
```

```
sudo apt install maven -y
```

```
java -version
```

```
mvn -version
```

```
ubuntu@Build-server:~$ java --version
openjdk 17.0.18 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-1, mixed mode, sharing)
ubuntu@Build-server:~$ mvn --version
Apache Maven 3.9.12
Maven home: /usr/share/maven
Java version: 17.0.18, vendor: Ubuntu, runtime: /usr/lib/jvm/java-17-openjdk-amd64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "7.0.0-1004-aws", arch: "amd64", family: "unix"
ubuntu@Build-server:~$
```

Step 3: Clone GitHub Repository

```
git clone https://github.com/mrtechreddy/Java-Web-Calculator-App.git
cd Java-Web-Calculator-App
```

```
ubuntu@Build-server:~$ ls
Java-Web-Calculator-App  devops.pem
ubuntu@Build-server:~$ |
```

Step 4: Build Application

mvn clean package

```
[INFO]
[INFO] --- war:3.3.2:war (default-war) @ webapp-add ---
[INFO] Packaging webapp
[INFO] Assembling webapp [webapp-add] in [/home/ubuntu/Java-Web-Calculator-App/target/webapp-add-0.0.3]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/ubuntu/Java-Web-Calculator-App/src/main/webapp]
[INFO] Building war: /home/ubuntu/Java-Web-Calculator-App/target/webapp-add-0.0.3.war
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 3.614 s
[INFO] Finished at: 2026-05-17T10:00:48Z
[INFO]
```

WAR file generated inside target/ directory.

```
ubuntu@Build-server:~$ cd Java-Web-Calculator-App/
ubuntu@Build-server:~/Java-Web-Calculator-App$ ls
Jenkinsfile README.md pom.xml src target
ubuntu@Build-server:~/Java-Web-Calculator-App$ cd target/
ubuntu@Build-server:~/Java-Web-Calculator-App/target$ ls
classes                generated-test-sources  maven-status          test-classes           webapp-add-0.0.3.war
generated-sources      maven-archiver         surefire-reports      webapp-add-0.0.3
ubuntu@Build-server:~/Java-Web-Calculator-App/target$ |
```

Step 5: Configure pom.xml

Add Nexus repository configuration under distributionManagement.

```
<distributionManagement>
  <repository>
    <id>maven-releases</id>
    <url>http://3.80.155.124:8081/repository/maven-releases/</url>
  </repository>
</distributionManagement>
</project>
```

Step 6: Configure Maven settings.xml

Add Nexus credentials in /etc/maven/settings.xml.

```
<!-- Nexus Authentication -->
<servers>

    <server>
<id>maven-releases</id>
<username>admin</username>
<password>admin123</password>
</server>

    <server>
<id>maven-snapshots</id>
<username>admin</username>
<password>admin</password>
</server>

</servers>

<!-- Mirrors -->
<mirrors>
```

Step 7: Deploy Artifact to Nexus

mvn deploy

```
[INFO]
[INFO] --- war:3.3.2:war (default-war) @ webapp-add ---
[INFO] Packaging webapp
[INFO] Assembling webapp [webapp-add] in [/home/ubuntu/Java-Web-Calculator-App/target/webapp-add-0.0.3]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/ubuntu/Java-Web-Calculator-App/src/main/webapp]
[INFO] Building war: /home/ubuntu/Java-Web-Calculator-App/target/webapp-add-0.0.3.war
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 3.614 s
[INFO] Finished at: 2026-05-17T10:00:48Z
[INFO]
```

2. Nexus Repository Server Setup

Step 1: Install Java

sudo apt update

sudo apt install openjdk-17-jdk -y

The screenshot displays the AWS Management Console interface for an EC2 instance. At the top, there's a search bar and navigation icons. The main content area shows the 'Instance summary' for the instance ID 'i-Oe83496df2823d81d', which is named 'Nexus-server'. The instance is in the 'Running' state, as indicated by the green checkmark icon. Key details include the public IPv4 address '3.80.59.36', the private IPv4 address '172.31.20.230', and the public DNS 'ec2-3-80-59-36.compute-1.amazonaws.com'. The instance was updated less than a minute ago.

```
ubuntu@Nexus-server:~$ java --version
openjdk 17.0.18 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-1, mixed mode, sharing)
ubuntu@Nexus-server:~$ |
```

Step 2: Download Nexus

```
14 wget https://download.sonatype.com/nexus/3/nexus-3.85.0-03-linux-x86_64.tar.gz
```

```
ubuntu@Nexus-server:~$ ls
nexus-3.85.0-03  nexus-3.85.0-03-linux-x86_64.tar.gz  sonatype-work
ubuntu@Nexus-server:~$ |
```

Step 3: Extract Nexus

tar -xvf latest-unix.tar.gz

Start Nexus

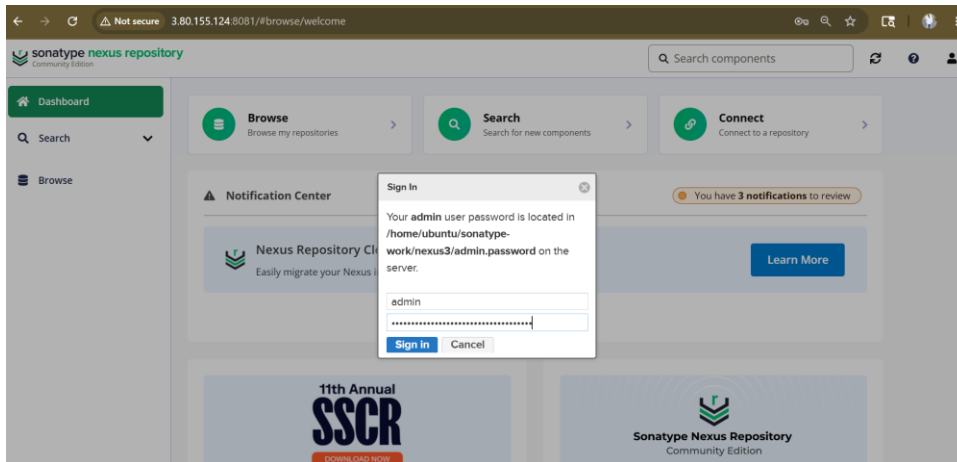
cd /opt/nexus/bin

./nexus start

```
16 tar -xvf nexus-3.85.0-03-linux-x86_64.tar.gz
17 ls
18 cd nexus-3.85.0-03/
19 ls
20 cd bin
21 ls
22 ./nexus
23 ./nexus start
24 ./nexus status
25 ./nexus status
```

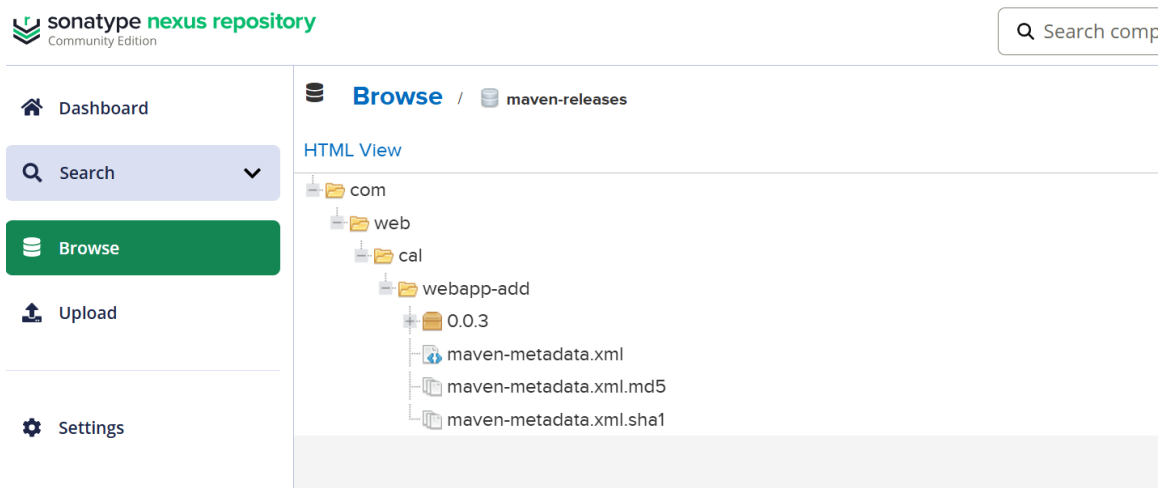
Step 5: Access Nexus Web Interface

http://3.80.59.36:8081



Step 6: Create Maven Hosted Repository

Create repository named maven-releases.

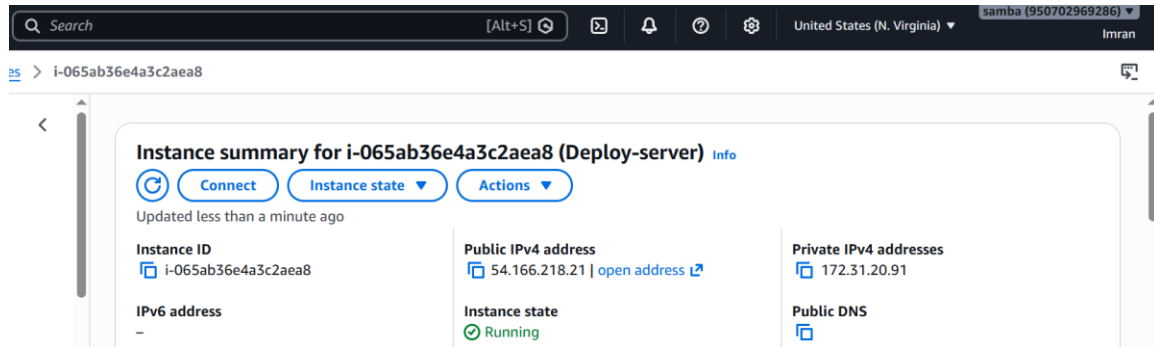


3. Deploy Server Setup

Step 1: Install Java and Apache Tomcat

```
sudo apt update
```

```
sudo apt install openjdk-17-jdk -y
```



```
ubuntu@Deploy-server:~$ java --version
openjdk 17.0.18 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-1, mixed mode, sharing)
ubuntu@Deploy-server:~$
```

Download Tomcat:

```
wget https://downloads.apache.org/tomcat/tomcat-9/v9.0.118/bin/apache-tomcat-9.0.118.tar.gz
```

```
8 wget https://dldn.apache.org/tomcat/tomcat-9/v9.0.118/bin/apache-tomcat-9.0.118.tar.gz
9 ls
10 tar -xvf apache-tomcat-9.0.118.tar.gz
```

Extract Tomcat:

```
tar -xvf apache-tomcat-9.0.118.tar.gz
```

```
apache-tomcat-9.0.118  apache-tomcat-9.0.118.tar.gz
ubuntu@Deploy-server:~$
```

Step 2: Start Tomcat

cd /opt/tomcat/bin

./startup.sh

```
ubuntu@Deploy-server:~$ cd apache-tomcat-9.0.118/
ubuntu@Deploy-server:~/apache-tomcat-9.0.118$ ls
BUILDING.txt  LICENSE  README.md  RUNNING.txt  conf  logs  webapps
CONTRIBUTING.md  NOTICE  RELEASE-NOTES  bin  lib  temp  work
ubuntu@Deploy-server:~/apache-tomcat-9.0.118$ cd bin
ubuntu@Deploy-server:~/apache-tomcat-9.0.118/bin$ ls
bootstrap.jar  ciphers.sh  commons-daemon-native.tar.gz  daemon.sh  setclasspath.bat  startup.sh  version
catalina-tasks.xml  commons-daemon.jar  digest.bat  setclasspath.sh  tomcat-juli.jar  version
catalina.bat  configtest.bat  makebase.bat  shutdown.bat  tomcat-native.tar.gz
ciphers.bat  configtest.sh  makebase.sh  startup.bat  tool-wrapper.bat
tool-wrapper.sh
ubuntu@Deploy-server:~/apache-tomcat-9.0.118/bin$ ./startup.sh
Using CATALINA_BASE:   /home/ubuntu/apache-tomcat-9.0.118
Using CATALINA_HOME:   /home/ubuntu/apache-tomcat-9.0.118
Using CATALINA_TMPDIR: /home/ubuntu/apache-tomcat-9.0.118/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /home/ubuntu/apache-tomcat-9.0.118/bin/bootstrap.jar:/home/ubuntu/apache-tomcat-9.0.118/b
t-juli.jar
Using CATALINA_OPTS:
Tomcat started.
```

 Not secure 54.166.218.21:8080

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Apache Tomcat/9.0.118

If you're seeing this, you've successfully installed Tomcat. Co



Recommended Reading:
[Security Considerations How-To](#)
[Manager Application How-To](#)
[Clustering/Session Replication How-To](#)

Developer Quick Start
[Tomcat Setup](#) [Realms & AAA](#) [Examples](#)
[First Web Application](#) [JDBC DataSources](#)

Step 3: Copy WAR File to Deploy Server

```
126 scp -i /home/ubuntu/devops.pem Java-Web-Calculator-App/target/webapp-add-0.0.3.war ubuntu@54.160.139.225:/home/ub
rtu/apache-tomcat-9.0.118/webapps/
```

Step 4: Access Application
http://54.166.218.21:8080

Addition

5

Calculator

first number:

Second number :

☒ addition
☐ subtraction
☐ production

Issues Encountered and Fixes

1. Permission denied for PEM file → Fixed using `chmod 400 devops.pem`
2. Maven deployment failed → Corrected Nexus repository ID and credentials
3. Tomcat startup issue → Verified Java installation and Tomcat logs
4. SCP connection issue → Updated security groups and PEM permissions

Conclusion

Successfully implemented a complete multi-server Java application deployment pipeline using Apache Maven, Nexus Repository Manager, and Apache Tomcat.